

Who considers, who owns? Multi-study evidence on the behavioral process of cryptocurrency adoption in the United States

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ABSTRACT

Cryptocurrency is a rapidly expanding landscape of digital assets spanning sociopolitical, psychological, technological, and financial domains. Although prior research identifies factors associated with owning cryptocurrency, most work overlooks the prerequisite considering stage. We draw on the Theory of Planned Behavior to examine whether individuals who would consider owning cryptocurrency differ from owners and those who reject the possibility of owning outright. Using three independent, nationally representative samples of US adults ($n=803$; $n=1908$; $n=2086$) and a modeling approach that allows us to compare the three groups, we find that roughly one in five respondents do not own, but would consider owning, cryptocurrency. While both considerers and owners share similar demographics and personality traits, considerers are not like owners in all ways or to the same degree: Conservative political ideology and technological optimism (support for developing artificial intelligence) relate more strongly to considering than owning, whereas stock ownership and disruptive orientations (the “Need for Chaos”) relate more strongly to owning than considering. Our data also mirror aggregate market patterns, with most respondents considering and owning Bitcoin (followed by Ethereum and other cryptocurrencies, e.g., Dogecoin and XRP). However, we further demonstrate that many individuals would consider or own more than one type. Taken together, the implication is that focusing only on ownership or certain types of cryptocurrency may obscure meaningful differences in evaluation and behavior. Accounting for these differences is therefore essential for understanding what cryptocurrency represents to various individuals, as well as effective communication, segmentation, and policy design.

1. Introduction

Since the emergence of Bitcoin in 2008, the cryptocurrency landscape has rapidly evolved amid a torrent of digital assets—more than 31 million coins and tokens collectively valued at nearly 2.5 trillion US dollars as of February 2026 (CoinMarketCap, 2026).¹ While two coins, Bitcoin and Ethereum, consistently account for nearly three-quarters of the total market capitalization, a growing set of alternative cryptocurrencies make up the rest of the sector, varying widely in purpose, viability, and price. Previous research on adoption patterns in the United States suggests that between 14% and 30% of US adults own or have

owned cryptocurrency at some point, and that owners differ from non-owners across sociopolitical, psychological, technological, and financial domains (see, e.g., Faverio et al., 2023; Ferguson et al., 2023; Littrell et al., 2024). In this sense, cryptocurrencies vary not only in their value and market reach, but also in the range of potential factors that drive their use and appeal.

Importantly, such individual-level heterogeneity mirrors a broader pattern in behavioral research, where a similar action can stem from diverse and even contradictory motivations, e.g., perceived utility (Davis, 1989), self-identity (Escalas and Bettman, 2005), status appeal or exclusivity (O’Cass and McEwen, 2006), and control (Venkatesh et al.,

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¹ The total market capitalization was approximately 2.3 trillion US dollars when we began collecting our first sample (in October 2024), 3.2 trillion when we began fielding Study 2 (in February 2025), and 3.3 trillion when we began fielding Study 3 (in June 2025). The number of cryptocurrencies at each point was approximately 1.7 million, 8.6 million, and 16.4 million, respectively. While thousands of cryptocurrencies are created daily (see, e.g., CoinMarketCap, 2026), many coins and tokens have little to no value and/or are inactive (see, e.g., Lee, 2026). Thus, we reference the total counts of cryptocurrencies and market values to illustrate the scope and proliferation of cryptocurrency, not to suggest parity of all cryptocurrency types.

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2003). Yet, most extant literature focuses only on existing cryptocurrency owners—an approach that risks conflating the factors associated with end-stage behavior with those associated with the prerequisite considering stage (see, e.g., Karahanna et al., 1999). Attitude-behavior theories like the Theory of Planned Behavior (TPB) posit that considering precedes intention and intention precedes taking action, but neither considering nor intention guarantee behavioral follow-through (Ajzen, 1985, 2020; Fishbein and Ajzen, 1975). Combined with other limitations (e.g., selective samples and inadequate modeling strategies), prior research offers a fragmented and potentially incomplete picture of who might engage with cryptocurrency (or not) and why.

Accordingly, we ask: Which factors are associated with *considering* cryptocurrency adoption, as similar to or distinct from those associated with rejecting and owning? Drawing on the TPB, we conceptualize cryptocurrency adoption as a behavioral process and assess whether “considerers” are a meaningful group. We argue that, if it is the case that “background factors” indirectly relate to both intentions and behavior via considerations (Ajzen, 2020), then an explicit measure of considering may help to capture whether there are differences among those who are open to the possibility of adopting cryptocurrency, regardless of whether intention or action occur.

We examine whether “considerers” (i.e., those who do not own, but would consider owning) are similar and/or distinct from rejectors (i.e., those who do not own, and would not consider owning) and existing owners using three large, nationally representative, cross-sectional samples of US adults ($n = 803$; $n = 1908$; $n = 2086$) and an analytic approach that allows for simultaneous evaluation of these groups. Across the three studies, we consistently find that existing cryptocurrency owners *and* considerers are distinct from rejectors, but considerers are not like owners in all ways or to the same degree. For example, we find that being younger, male, and/or more open to experience increase the probability of both considering and owning cryptocurrency. However, more conservative ideology and technological optimism are more strongly associated with considering than owning, whereas stock ownership and disruptive orientations are more strongly associated with owning than considering.

Taken together, our findings align with foundational behavioral theories to suggest that the factors associated with considering a behavior are not necessarily the same as those associated with the behavior itself. Our findings underscore the importance of measuring and modeling intermediate stages of behavior, particularly for anticipating who may be more likely to engage with cryptocurrency before adoption occurs.

2. Existing evidence on cryptocurrency adoption

The origins of cryptocurrency adoption stem from the Bitcoin white paper, published under the pseudonym Satoshi Nakamoto, in 2008. Having emphasized the pitfalls of relying on centralized financial systems, Nakamoto introduced the first-mover cryptocurrency, Bitcoin—a peer-to-peer digital monetary network designed to operate sans third-party monetary institutions (Nakamoto, 2008; see also Bailey et al., 2021; 2024). This ethos continues to shape scholarship on Bitcoin and other cryptocurrency behavior, much of which assumes that this sort of decentralized monetary alternative would appeal to individuals who distrust conventional institutions, are open to emerging innovations, and are willing to reject the status quo (Ferguson et al., 2023).

Consistent with this view, prior evidence finds that cryptocurrency owners are different from non-owners across multiple domains: Demographically, owners tend to be younger, male, non-white and/or Hispanic, and have higher levels of education, income, and religiosity (Ferguson et al., 2023; Littrell et al., 2024). With regard to personality, owners are often more open to experience, less conscientious, and more risk-tolerant than non-owners (Ferguson et al., 2023). While evidence is mixed as to whether ownership is associated with conventional left-/right political orientations (Littrell et al., 2024), some data suggest that

ownership aligns with more conservative attitudes regarding what the government should or should not do (Ferguson et al., 2023; see also Ellis and Stimson, 2012; Gamble, 2013).

Emerging work also highlights potential differences in personality adaptations, or context-specific traits. For example, owners often score lower on authoritarianism, in line with the notion that they may be less concerned with conformity and more open to social change (Littrell et al., 2024; see also Engelhardt et al., 2023). Owners also tend to score higher in anti-establishment orientations like Need for Chaos, i.e., the expressed desire to dismantle existing social and political systems amid periods of real and perceived marginalization (Petersen et al., 2023). Willingness to take financial risks and other investment behaviors are also associated with ownership (Bohr and Bashir, 2014; Robba et al., 2024), suggesting adoption may depend on how one thinks about cryptocurrency in terms of risk and their ability to follow through with investment decisions.

2.1. Considering cryptocurrency

Despite these insights, most existing research focuses only on those who own cryptocurrency—an approach that, we argue, risks conflating the factors associated with adoption with those related to the prerequisite considering stage (see, e.g., Karahanna et al., 1999). According to the Theory of Planned Behavior (TPB), intentions are a necessary but insufficient condition for behavior: Considerations about a behavior precede intentions and intentions precede action, but neither considerations nor intentions guarantee behavior will occur (Ajzen, 1985, 2020; Fishbein and Ajzen, 1975).² Bosnjak et al. (2020) describe the TPB thusly:

“... [H]uman behavior is guided by three kinds of considerations: beliefs about the likely consequences of the behavior (behavioral beliefs), beliefs about the normative expectations of others (normative beliefs), and beliefs about the presence of factors that may facilitate or impede performance of the behavior (control beliefs). In their respective aggregates, behavioral beliefs produce a favorable or unfavorable attitude toward the behavior; normative beliefs result in perceived social pressure or subjective norm; and control beliefs give rise to perceived behavioral control or self-efficacy. The effects of attitude toward the behavior and subjective norm on intention are moderated by perception of behavioral control. As a general rule, the more favorable the attitude and subjective norm, and the greater the perceived control, the stronger should be the person’s intention to perform the behavior in question. Finally, given a sufficient degree of actual control over the behavior, people are expected to carry out their intentions when the opportunity arises. Intention is thus assumed to be the immediate antecedent of behavior. To the extent that perceived behavioral control is veridical, it can serve as a proxy for actual control and contribute to the prediction of the behavior in question.” (353).

Furthermore, the TPB posits that “background factors” (i.e., “[p]ersonality traits, intelligence, demographic characteristics, life values, and other variables of this kind,” see Ajzen, 2020, 318) are situated *prior* to behavioral, normative, and control beliefs (Ajzen, 2020). This

² According to Ajzen (2020), the Theory of Planned Behavior (TPB) was developed from the Theory of Reasoned Action (TRA), and the distinction between them lies in objective and subjective control: “[T]he theory of reasoned action is a special case of the TPB. The major difference between the two theories is that the TPB includes actual and perceived behavioral control as additional determinants of intentions and behavior. When people have perfect volitional control over the behavior of interest, and when they strongly believe that they are capable of performing the behavior if they so desire, behavioral control is irrelevant, and the TPB reduces to the theory of reasoned action[...]” (316).

suggests that studies concerned primarily with owning cryptocurrency can only provide insights into which factors relate to *following through* with adoption—not why individuals might consider (or not) adopting in the first place (Ajzen, 2020). For example, an individual who is more open to adopting cryptocurrency may refrain from taking action if they are uncertain about market volatility or lack the resources to invest. Conversely, an individual who is not initially open to adopting cryptocurrency may eventually take action if faced with a financial opportunity or persuasive social cues. We present a schematic diagram of the TPB, modified to include background factors and in the context of cryptocurrency ownership, in Fig. 1.

2.2. The present study

Accordingly, we ask: Which factors are associated with *considering* cryptocurrency adoption, as similar to or distinct from those associated with rejecting and owning? To reiterate, prior research implicitly assumes that, because owners have taken action, these individuals are meaningfully different from non-owners. Yet, by not accounting for whether non-owners would consider owning cryptocurrency, researchers overlook the fact that considerations and intentions are necessary conditions for behavior (Ajzen, 1985, 2020; Fishbein and Ajzen, 1975). The TPB would further contend that, in the context of cryptocurrency adoption, all owners were once considerers but not all considerers become owners (Ajzen, 1985, 2020; Fishbein and Ajzen, 1975). By this view, we might expect considerers to share at least some similarities with owners, though not necessarily in all ways or to the same degree.

We argue that, if it is the case that considerers share similarities and differences with both rejectors *and* owners, then ownership-focused work risks providing a fragmented and potentially incomplete picture of who might engage with cryptocurrency (or not) and why. In turn, we account for “considerers” and empirically test whether they are a meaningful group. To foreshadow our methodological approach (discussed in detail below), we operationalize cryptocurrency engagement as an ordinal variable and use generalized ordered logit, or partial proportional odds, models to examine whether independent variables elucidated by previous ownership research are consistent for both considerers and owners, or vary in strength.

Before proceeding, it is important to note that we do not aim to explicitly test the mediation pathways proposed by the TPB; rather, we leverage the TPB to test whether considerers are similar to and/or different to those who reject and own. By modeling the relationship between background factors and rejection (a lack of willingness), consideration (openness to a behavior), and ownership (realized behavior) simultaneously, our approach helps to overcome common limitations of previous research.³ If considerers are a meaningful group, then our findings should help to expand our understanding of the individual differences associated with cryptocurrency engagement and provide insights into potential market growth. For example, if considerers are in some ways similar to rejectors, then this might suggest that the barriers to future adoption lie in the maintenance of unfavorable attitudes, norms, and control beliefs. However, if considerers are in some ways similar to owners, then this might suggest who is more likely to take action upon developing more favorable control beliefs. Lastly, if considerers are in some ways distinct from both rejectors and owners,

then this distinction may represent an untapped segment of potential adopters, offering clues into how cryptocurrency adoption might stall and/or continue to grow.

3. Data and measures

To test which factors are associated with considering cryptocurrency, as similar to or distinct from those associated with rejecting or owning, we collected three original, nationally representative, cross-sectional samples of non-institutionalized US adults (aged 18+).⁴ We conducted Study 1 ($n = 831$) via the Cooperative Election Study (CES) between October 1, 2024 and December 10, 2024.⁵ We conducted Study 2 ($n = 1932$) via Lucid between February 14 and March 1, 2025. We conducted Study 3 ($n = 2128$) via Verasight between June 10 and June 14, 2025. To be explicit, all three studies are designed to be representative of US adults and include the same measures; however, each study draws on an independent, non-overlapping sample collected from different providers at different points in time. This design is intentional, as we aim to assess whether results are consistent across samples by treating each study as an independent test.

In the interest of space, we include all details on these data in Appendix A of Supplementary Material and summarize the source, protocol, dates fielded, and sample sizes for each study in Table 1. While the total n represents all consenting respondents in each sample, the analytic n represents the number of respondents included in our analysis after accounting for attrition and missing values on key variables. In all three studies, the analytic n includes between 97 and 99% of the total n , yielding samples of $n = 803$ for Study 1, $n = 1908$ for Study 2, and $n = 2086$ for Study 3. To adjust for any residual differences in each sample, all analyses account for sampling weights.

It is worth noting that, even though all three datasets are nationally representative, our net samples differ somewhat in their median age category (50 - 64 in Study 1, 45 - 64 in Study 2, and 30 - 49 in Study 3). Such differences can arise even in representative samples due to missing values and survey context; for example, we might expect that older respondents are more likely to participate in a post-election survey such as that used for Study 1 (DeBell and Krosnick, 2009). Given prior research consistently finds that age is a key correlate with cryptocurrency ownership in the US, independent of other effects (see, e.g., Faverio et al., 2023; Ferguson et al., 2023; Littrell et al., 2024), we argue that the differences in age distributions across samples are not a methodological limitation—rather, they provide a natural stress test for our results. As demonstrated in the results that follow, our findings are largely consistent across samples, and we later discuss the implications of sample differences in age for observed ownership rates.

3.1. Target population

We focus on US adults for both theoretical and empirical reasons: Available evidence suggests that the US is currently one of the most active digital asset markets (CoinMarketCap, 2026) and much of the prior research on cryptocurrency adoption (e.g., Faverio et al., 2023; Ferguson et al., 2023; Littrell et al., 2024) has been in the US context. The regulatory environment and increasing politicization surrounding

³ While we are not the first to apply the TPB to the context of cryptocurrency adoption, prior work has several limitations. For example, research that accounts for interest and/or intentions for adoption (see, e.g., Martin et al., 2022; Stanciu et al., 2024) helps to move beyond ownership, but nevertheless assumes that a meaningful proportion of individuals have favorable and sufficiently strong attitudes, norms, and control beliefs (Ajzen, 2020). Others model outcomes like intention or awareness separately (Stanciu et al., 2024), such that they cannot compare potential differences between groups.

⁴ Prior to participating, study respondents provided informed consent. All studies were deemed exempt from formal review by the Colorado State University Institutional Review Board (IRB) and were carried out in accordance with the American Political Science Association (APSA) Principles and Guidance on Human Subject Research.

⁵ As discussed in Appendix A in Supplementary Material, Study 1 included two waves of data collection, with 1000 respondents included in Wave 1 and 831 respondents retained in Wave 2. Because all cryptocurrency-related measures for our dependent variable were included in Wave 2, we treat the Wave 2 sample ($n = 831$) as the total baseline sample.

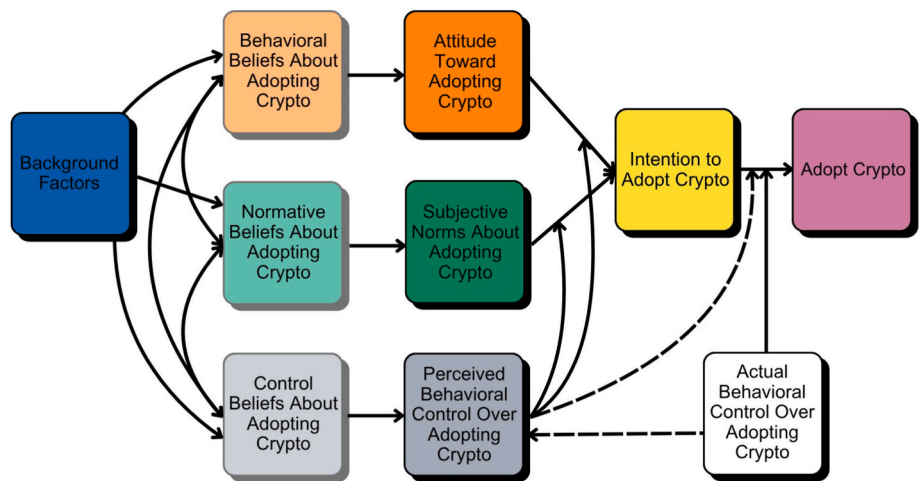


Fig. 1. Schematic diagram of the Theory of Planned Behavior (TPB), modified to include background factors and in the context of cryptocurrency ownership, adapted from Ajzen (2019).

Table 1
Summary of studies.

Study	Source	Protocol	Dates Fielded	Total n	Analytic n
1	Cooperative Election Study	Online probability sample matched to the target population on voter registration status, age, race, gender, and education	October 1 -December 10, 2024	831	803
2	Lucid	Online non-probability sample matched to the target population on age, gender, ethnicity, and region	February 14 -March 1, 2025	1932	1908
3	Verasight	Online multi-mode sample matched to the target population on age, gender, ethnicity, income, education, region, and partisanship	June 10 – June 14, 2025	2128	2086

Note: Sample source, sampling protocol, dates fielded, total *n*, and analytic *n* for Studies 1-3. Total *n* represents the number of respondents who consented to participate in each study. Analytic *n* represents the number of respondents included in our analysis after accounting for attrition, missing values on key variables, and failed attention checks. The target population for all samples is non-institutionalized US adults (aged 18+).

digital assets also creates conditions under which the individual differences examined in previous studies may be especially salient to US consumers (see, e.g., Congressional Digest, 2025; EO 11789, 2025; Perkins, 2020; Tierno, 2023; 2025). Moreover, focusing on the US allows us to use previously validated measures (e.g., trust in the federal government and policy support) that may not translate into cross-national contexts or other domains, enabling us to be more confident in the validity of our findings and how they might fit in with past and future work.

3.2. Dependent variable

To examine who in our sample owns, or would consider owning, cryptocurrency, we asked respondents in all three studies the following question, adapted from previous research (Ferguson et al., 2023): “Do you personally, or jointly with a spouse, own any type of cryptocurrency, such as Bitcoin or Ethereum?” We branched respondents who selected *Yes* into: Which type(s) of cryptocurrency do you own? Select all that apply. *Bitcoin, Ethereum, Something Else (open-ended)*. We branched respondents who selected *No* to the ownership question into: Would you ever consider owning any type of cryptocurrency, such as Bitcoin or Ethereum? We branched respondents who selected *Yes* into: Which type(s) of cryptocurrency would you consider owning? Select all that apply. *Bitcoin, Ethereum, Something Else (open-ended)*.⁶

Using the “Which type(s)” questions as additional check, we found nine respondents in Study 1, 27 respondents in Study 2, and 15 respondents in Study 3 whose open-ended responses for *Something Else* (e.g., “None” or “Sorry, I don’t”) indicated they do not own cryptocurrency. We also identified two respondents in Study 2 whose open-ended responses for *Something Else* (i.e., “None”) indicated they would not consider owning cryptocurrency. After ensuring that these respondents did not select other types of cryptocurrency, we recategorized these respondents as non-owners/non-considerers in their respective sample.

From these responses, we created a three-point variable categorizing respondents as: 1) does not own and would not consider, 2) does not own but would consider, or 3) owns. While we consider this to be an ordinal measure reflecting the process of behavior, we acknowledge that this conceptualization may raise concerns among some readers, given the sequence of our survey questions. However, we emphasize that there is no other valid or more straightforward way, at least to our knowledge, to distinguish between owners and non-owners in this context. Designing the survey such that it first identified owners, then identified who among non-owners would consider owning, allows us to clearly distinguish between these groups—a task that, we argue, would be more challenging with other approaches that risk introducing ambiguity (e.g., asking respondents whether they would consider prior to owning, or asking about owning and considering in a single question). Even if our survey introduced non-owning respondents to the concept of cryptocurrency for the first time, our chosen sequence of questions should nevertheless provide insights into who may or may not oppose the possibility of owning cryptocurrency altogether.

⁶ For brevity, we use the terms “would not consider” and “rejectors” interchangeably.

3.3. Independent variables

As previously discussed, prior research has identified a wide range of individual differences associated with owning cryptocurrency and we expect our own findings for ownership to largely align with this work. However, given the theoretical framework and multidimensional nature of cryptocurrency engagement, it is neither theoretically nor empirically straightforward to make assumptions about which factors are associated specifically with each behavioral stage. Not only does the TPB emphasize the *process* of behavior over background factors, but the latter are virtually impossible to parse out in terms of how they might interact with attitudes, norms, and control beliefs (see again Fig. 1). As such, we rely on prior literature as a general guide for which variables to include in our analysis but allow the data to reveal how these factors operate for rejecting, considering, and/or owning groups.

We do, however, propose three alternative hypotheses based on the broader possibilities outlined in the TPB. Importantly, these hypotheses concern the substantive role of considering as it relates (or not) to rejecting and/or owning, rather than any specific independent variable. We do not assume that a single hypothesis characterizes all variables; instead, we expect to find that different factors align with different hypotheses—consistent with the idea that considerers may share similarities and differences with both rejectors and owners, but not in all ways or to the same degree. We test these proposed alternatives against three null hypotheses: The first underpins existing ownership literature, i.e., that owners are not distinct from non-owners, including considerers. The second is implied by the measurement strategy in prior ownership literature and motivates the present study, i.e., that considerers are not distinct from rejectors. The third proposes that, to the extent that we reject the second null hypothesis (which would suggest that considerers are distinct from rejectors), then they are like owners in all ways and to the same degree. The three alternative hypotheses are as follows:

- H1. Owners are distinct from all non-owners, including considerers.
- H2. Despite both being non-owners, considerers are distinct from rejectors.
- H3. To the extent that considerers are distinct from rejectors, considerers are not like owners in all ways or to the same degree.

We summarize each independent variable below. Additional details (including question wording and coding, descriptive statistics, and pairwise correlations) are included in [Appendices B - D](#) in Supplementary Material.

3.3.1. Sociodemographics

Prior research has found that cryptocurrency owners are demographically younger, male, non-white and/or Hispanic, higher-income, higher-education, and more religious, compared to non-owners (Ferguson et al., 2023; Littrell et al., 2024). Thus, we also include measures for Age, Male gender, White race, Hispanic ethnicity, Education level, household Income, and Religiosity.

3.3.2. Personality

Extant literature also proposed that, given its novel and decentralized nature, cryptocurrency might appeal to individuals with specific personality traits. Ferguson et al. (2023), for example, proposed that owners may be more open to engaging with new ideas, experiences, and beliefs (i.e., openness), less concerned with organization and self-control (i.e., conscientiousness), more outgoing and enjoying of socialization (i.e., extraversion), more cooperative towards others (i.e., agreeableness), and less concerned with managing their environment (i.e., neuroticism, see Ferguson et al., 2023; see also Costa and McCrae, 1992). We might also expect owners to express a stronger Need for Cognition, i.e., the “tendency to engage in and enjoy effortful cognitive activity” (Lins de Holanda Coelho et al., 2020, 1870; see also Cacioppo and Petty, 1982). Thus, like previous research, we account for Big Five personality traits

(Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism) and a composite measure of Need for Cognition ($\alpha_{\text{Study1}} = 0.81$; $\alpha_{\text{Study2}} = 0.71$; $\alpha_{\text{Study3}} = 0.68$).

3.3.3. Political orientation

Recall that previous research found some evidence that conservative policy preferences are associated with owning cryptocurrency (Ferguson et al., 2023; see also Ellis and Stimson, 2012; Gamble, 2013). In contrast to measures like party identification (which reflect the direction and strength of one's psychological attachment to a major political party, see Campbell et al., 1960) and symbolic ideology (which merely captures one's self placement on a seven-point, liberal-to-conservative scale, see Ellis and Stimson, 2012), operational ideology averages together individuals' policy preferences across a range of political issues, providing a more precise and latent indicator of political views (Ellis and Stimson, 2012). For example, whereas one might identify as independent or moderate in terms of partisanship or symbolic ideology, they might express more consistently liberal or conservative political attitudes.

We argue such discrepancies may be especially important for understanding those who engage with cryptocurrency, as these individuals may hold distinct political preferences despite rejecting conventional political identities and institutional norms. Therefore, we include a measure of *Operational Ideology* that averages together respondents' support or opposition to four policy proposals: 1) Banning assault rifles, 2) Building the US/Mexico border wall, 3) Allowing a woman the right to abortion as a matter of choice, and 4) Increasing fossil fuel production in the US. Prior to scaling these items together, each item was coded such that lower values represent more liberal preferences and higher values represent more conservative preferences. Item and factor analyses confirm these four items fit together well into a unidimensional *Operational Ideology* scale ($\alpha_{\text{Study1}} = 0.77$; $\alpha_{\text{Study2}} = 0.58$; $\alpha_{\text{Study3}} = 0.54$).

3.3.4. Personality adaptations

Because previous research found that anti-establishment orientations are associated with owning cryptocurrency, we also account for respondents' Need for Chaos, a previously validated measure capturing one's disaffection with society and the political system independent of partisanship, ideology, and beliefs (Petersen et al., 2023; see also Farhart et al., 2023). Although *Need for Chaos* is positively associated with anti-establishment orientations included in previous cryptocurrency adoption research (e.g., the Dark Triad, Dark Tetrad, and conspiracy thinking, see Ferguson et al., 2023; Littrell et al., 2024; Martin et al., 2022), it uniquely “emerges in an interplay between status-oriented personality traits and social contexts of real and perceived marginalization” (Petersen et al., 2023, 20, p. 1486). That *Need for Chaos* helps to capture a combination of state and trait suggests it may be a more appropriate measure of the extent to which certain individuals are willing to challenge the status quo (Littrell et al., 2024).⁷

To measure Need for Chaos, respondents were asked to indicate the extent to which they agree with seven statements, including “When I think about our political and social institutions, I cannot help thinking ‘just let them all burn’” and “We cannot fix the problems in our social institutions, we need to tear them down and start over” (Petersen et al., 2023). As is convention, we combined responses into a seven-item *Need for Chaos* scale ($\alpha_{\text{Study1}} = 0.83$; $\alpha_{\text{Study2}} = 0.92$; $\alpha_{\text{Study3}} = 0.92$).

Previous work also found a negative relationship between authoritarianism and ownership; thus, we account for *Authoritarianism*, another personality adaptation that can vary in response “to perceive[d] threat in social change and challenges to social cohesion”

⁷ Using data from Study 2, we tested whether measures included in the Dark Triad (i.e., narcissism, Machiavellianism, and psychopathy) and conspiracy thinking were associated with considering and owning. Similar to Ferguson et al. (2023), we find that these traits are positively associated with both stages of the adoption process but ultimately non-significant in multivariate models.

(Engelhardt et al., 2023, 540), using a four-item scale ($\alpha_{\text{Study1}} = 0.70$; $\alpha_{\text{Study2}} = 0.56$; $\alpha_{\text{Study3}} = 0.54$) that tasks respondents with choosing between pairs of qualities they deem desirable for children.

3.3.5. Trust

Existing research (e.g., Littrell et al., 2024) further indicates that ownership is positively associated with trust in government and trust in people—another potential signal that those who engage with cryptocurrency have different judgments and responses to social and political contexts. Thus, we also account for *Trust Government* and *Trust People*. We also account for *Trust Banks*, in case the relationship between trust and cryptocurrency engagement relates more to how people think about financial institutions.

3.3.6. Technological optimism

Prior research on technological innovation adoption suggests individuals' evaluations of one innovation can help to inform evaluations of other technology, shaping how they perceive broader risks and rewards (Greenhalgh et al., 2005). In this sense, assessing one's attitudes towards other technology may help to differentiate those who are predisposed to engaging with emerging, potentially disruptive, innovations from those who are skeptical or otherwise risk-averse. Accounting for individual differences in attitudes toward other technology may also help us to better capture domain-specific orientations and avoid conflating enthusiasm for novel technology with cryptocurrency-specific motivations.

Our measure of technological optimism, *Supports Developing AI*, indicates whether respondents support or oppose developing artificial intelligence (AI) for everyday use. Importantly, this question was included alongside other policy issues in our surveys, but results from item and factor analyses indicate that *Supports Developing AI* does not fit together with other items in our *Operational Ideology* scale.

3.3.7. Other investments and risk

Because previous studies found that stock ownership and risk-taking are positively associated with ownership (e.g., Ferguson et al., 2023; Robba et al., 2024), we include a measure of *Stock Owner* to indicate whether respondents report owning any stocks. We also include a measure of *Investing Risk* to account for individuals' willingness to make risky investments, which uses a three-item measure ($\alpha_{\text{Study1}} = 0.70$; $\alpha_{\text{Study2}} = 0.82$; $\alpha_{\text{Study3}} = 0.82$) from the Domain-Specific Risk-Taking Scale (i.e., DOSPERT, see Weber et al., 2002). For each item included in *Investing Risk*, respondents were asked to indicate the likelihood they “would engage in the described activity or behavior if you were to find yourself in that situation.” In contrast to our measure of *Stock Owner*, asking respondents about how likely they would be to “Invest[] 10% of your annual income in a moderate growth mutual fund” and “Invest[] 5% of your annual income in a very speculative stock” allows us to effectively examine who might engage in these activities, if given the opportunity, regardless of their financial status or behaviors.

4. Results

To organize our empirical strategy and link it to our theoretical framework, we present our findings in four parts: Because our research question and hypotheses rest on the assumption that a meaningful share of non-owning individuals would consider adopting cryptocurrency, we first examine baseline rates of considering and owning in all three independent samples. Second, we examine the types of cryptocurrencies that respondents report considering or owning, as doing so may provide descriptive insights into how considerers relate to the rejecting and/or owning groups. Third, we assess results from a series of logit models that test whether there is indeed a distinction between cryptocurrency

owners and all non-owners, as established in previous literature (H1); we also estimate a series of models contrasting considerers with rejectors to test whether the non-owning category is an internally homogeneous group (H2). Fourth, we evaluate results from a series of multivariate models that compare all three groups simultaneously, such that we can determine whether there are measurable differences between rejecting, considering, and owning groups (H3). We present a diagram of our analytic strategy in Fig. 2.

4.1. Finding 1: rates of rejecting, considering, and owning

We begin by examining the rates of rejecting, considering, and owning cryptocurrency in each independent sample. Consistent with previous estimates (Faverio et al., 2023; Ferguson et al., 2023), approximately 13% of our sample in Study 1 and 18% of our sample in Study 2 report owning cryptocurrency (see Fig. 3). The proportion of owners in Study 3 is markedly higher (approximately 32%) but still in line with evidence indicating that roughly a third of Americans have owned or used cryptocurrency at some point (Littrell et al., 2024). We also find that roughly one-fifth of each sample would consider owning cryptocurrency (19% in Study 1, 20% in Study 2, and 22% in Study 3), leaving approximately 46% (in Study 1), 62% (in Study 2), and 68% (in Study 3) who report that they neither own or would consider owning. Overall, our data suggest that at least a plurality of respondents reject the possibility of owning cryptocurrency; however, “considerers” may constitute a real and potentially meaningful group.

4.2. Finding 2: types of cryptocurrencies considered and owned

We also present data on the types of cryptocurrencies considered or owned in Fig. 4. As shown in the left-hand panel of Fig. 4, most considerers (65%, 76%, and 67%, respectively) and owners (65%, 81%, and 51%, respectively) in all samples reported that they would consider or that they owned one type of cryptocurrency. A smaller percentage of considerers (35%, 24%, and 33%) and owners (35%, 19%, and 49%) reported more than one type.

As shown in the right-hand panel of Fig. 4, most considerers indicated that they would consider Bitcoin (93%, 89%, and 93%), followed by Ethereum (37%, 31%, and 37%), and “Something Else” (8%, 5%, and 4%). A similar pattern emerges among owners, with most indicating that they owned Bitcoin (70%, 76%, and 87%), followed by Ethereum (34%, 33%, and 53%), and “Something Else” (29%, 10%, and 9%). Importantly, these self-reported rates mirror aggregate market data (e.g., Bitcoin dominance trends, see CoinMarketCap, 2026), lending external

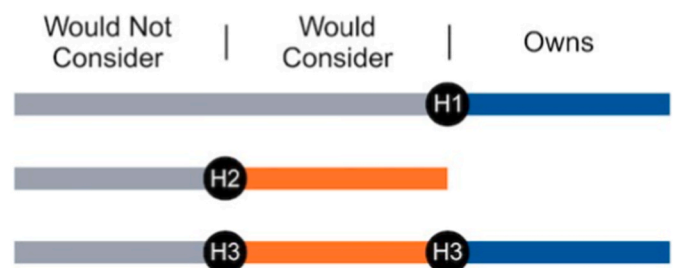


Fig. 2. Diagram of our analytic strategy based on categories of the dependent variable (would not consider, would consider, and owns cryptocurrency) and corresponding hypotheses. H1 tests whether owners differ from all non-owners, H2 tests whether considerers differ from rejectors, and H3 simultaneously tests whether there are measurable differences between rejecting, considering, and owning groups. Categories of the dependent variable are not necessarily representative of actual sample proportions discussed in section 4.1.

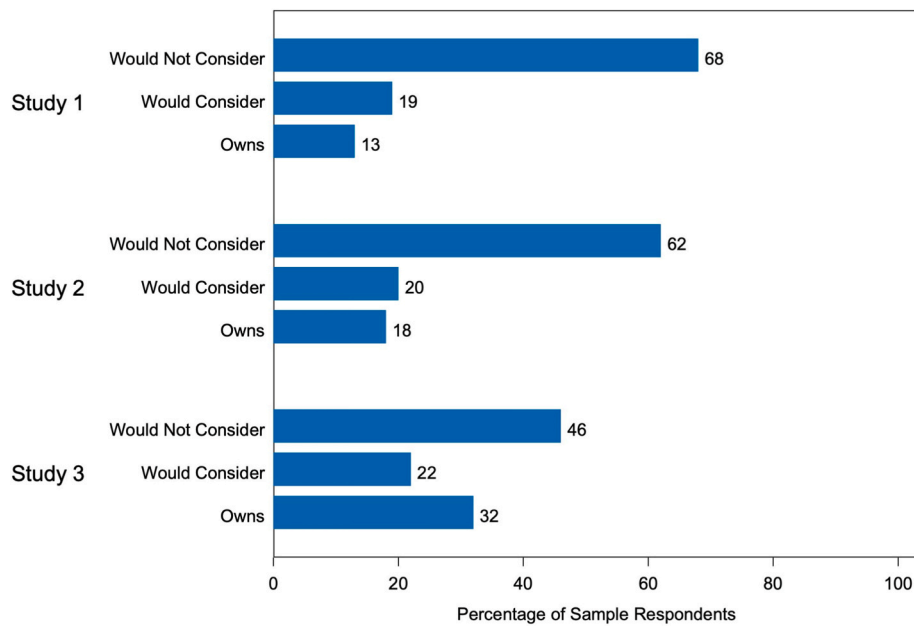


Fig. 3. Percentage of US adults in each independent sample who indicated that they would not consider owning cryptocurrency, do not own but would consider owning cryptocurrency, or own cryptocurrency.

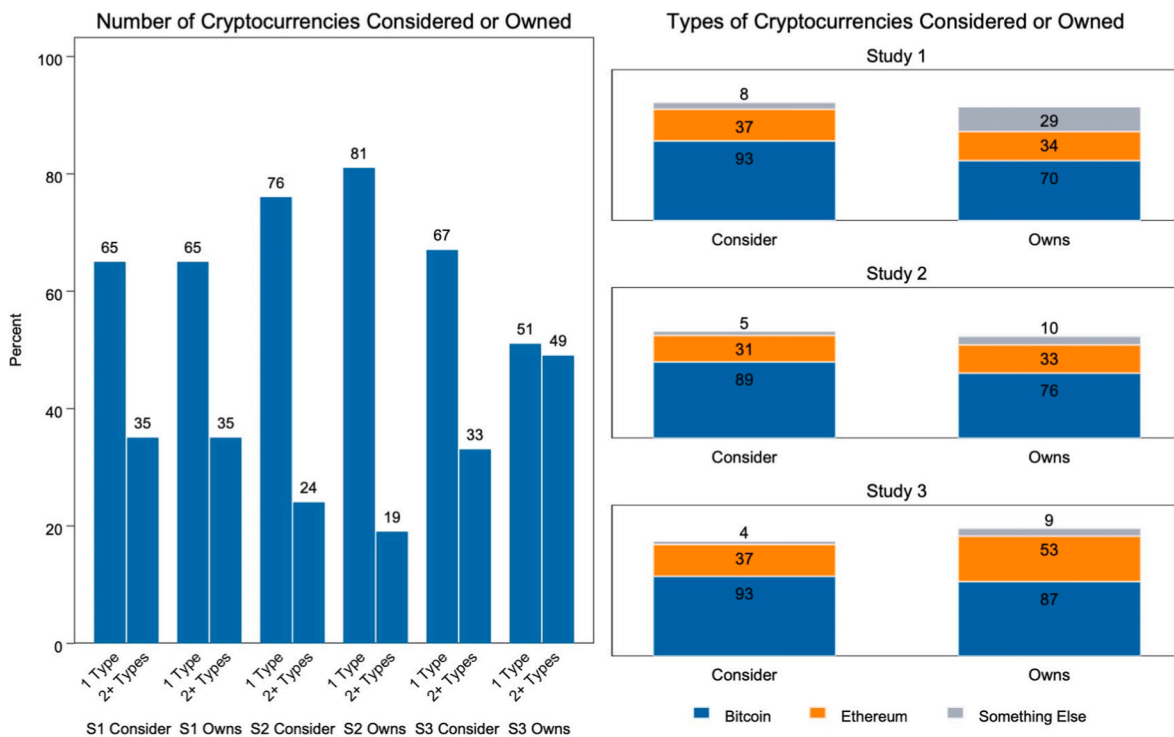


Fig. 4. Percentage of considerers and owners in each sample who indicated they would consider or owned 1 or 2+ types of cryptocurrency (left-hand panel). Percentage of considerers and owners by type of cryptocurrency considered or owned (right-hand panel).

validity to both our measures and the idea that considerers may be a meaningful group.⁸

Respondents who selected “Something Else” also received an

⁸ According to [CoinMarketCap \(2026\)](#), the relative market shares of Bitcoin, Ethereum, and other cryptocurrencies were approximately the following at the time of each study: 59%, 13%, and 28% (Study 1); 60%, 10%, and 30% (Study 2); 64%, 9%, and 27% (Study 3).

additional follow-up question about which other type(s) of cryptocurrency they would consider or owned. According to these responses, some non-owners view considering as a practical decision (e.g., “Whatever else makes sense as a speculation/investment” and “Any that is stable or rising in value”), whereas others expressed uncertainty (e.g., “I don’t really understand what it is” and “Not sure, I’d have to research it more”). While some considerers expressed a more general openness toward cryptocurrency (e.g., “I am open to learning about anything”), others were somewhat more explicit in their aversion to risk (e.g.,

“Whatever is proven and universally used” and “Something with less risk and expense”). As expected, and in contrast to considerers, owners were less ambiguous in how they described their retrospective choice, though some owners did not recall other types they own (e.g., “Don’t recall but it’s so minuscule ...” and “[B]its of a few other blockchain coins”).⁹ Taken together, these findings align with our expectations that considerers and owners may represent distinct psychological and behavioral groups.

4.3. Finding 3: owning vs. considering and considering vs. rejecting cryptocurrency

Having established that a sizeable share of respondents would consider owning cryptocurrency, and that considerers and owners may represent distinct psychological and behavioral groups, we now formally test our first two hypotheses: that owners are distinct from non-owners, including considerers (H1) and, despite both being non-owners, considerers are distinct from rejectors (H2). Critically, these tests allow us to determine whether our data replicate patterns in previous ownership studies and whether considerers constitute a distinct group. Moreover, if we do not find evidence for H1 or H2, there would be little justification for proceeding to test whether there are simultaneous differences between rejecting, considering, and owning groups (H3). We empirically test whether owners are distinct from non-owners, and whether considerers are distinct from rejectors, with a series of multivariate logit models.

For ease of interpretation, we plot these results in Fig. 5.¹⁰ We present these results in an order that reflects the cryptocurrency adoption process as informed by the Theory of Planned Behavior: the probability of “would not consider” versus “would consider” (H2) and the probability of not owning (“would not consider” combined with “consider”) versus owning (H1). While we cannot directly compare results between groups or across studies because they are estimated in separate models, organizing the results in this way allows us to look for consistent patterns and introduces the idea that these categories may follow an underlying ordinal progression.

Looking first to the probability of not owning versus owning (indicated by the blue circles and corresponding blue error bars), several independent variables align with patterns documented in extant literature (e.g., Bohr and Bashir, 2014; Ferguson et al., 2023; Littrell et al., 2024; Robba et al., 2024): Younger Age, higher Income, greater Trust Government, greater Investing Risk, greater Need for Chaos, and greater Investing Risk are directionally consistent in all samples and significantly associated with ownership in two or more samples. Coefficient estimates for Religiosity vary in direction across samples but are positive and significant in Study 2 and Study 3, consistent with prior work. Hispanic also appears sample dependent (significant and negative in Study 1 but significant and positive in Study 3), in line with previous evidence indicating positive or mixed effects. While other variables (Education, Openness, Conscientiousness, Agreeableness, Supports Developing AI) also demonstrate single-study effects, potentially speaking to the mixed or context-dependent patterns reported in previous work, the isolated relationship for Male (in Study 3) is more surprising given the strong and consistent gender gap observed in prior research. Regardless, we find strong support for the hypothesis that owners differ from non-owners, including considerers (H1).

Coefficient estimates for the probability of considering versus

rejecting (indicated by the orange circles and corresponding orange error bars) also support our hypothesis that non-owners are not necessarily a homogeneous group (H2). Here, younger Age, greater Openness, lesser Need for Cognition, greater (or more conservative) Operational Ideology, lesser Need for Chaos, greater Supports Developing AI, and greater Investing Risk distinguish considerers from rejectors in two or more samples. Notably, Male is also significant and positively associated with considering in two of three independent samples. Religiosity also varies for considerers across samples (significant and negative in Study 1 but significant and positive in Study 3), while Hispanic and Conscientiousness exhibit single-study effects. Overall, these results suggest that many of the same factors that distinguish owners from non-owners also distinguish considerers from rejectors; however, certain factors may operate differently for considering and owning groups. Because we cannot assess simultaneous effects across separate models for considering and owning, understanding whether and how these relationships differ requires a unified modeling approach.

4.4. Finding 4: differences between rejectors, considerers, and owners

Having replicated previous findings that owners are distinct from non-owners (H1) and further demonstrated that considerers are distinct from rejectors (H2), we now test our final hypothesis: To the extent that considerers are distinct from rejectors, considerers are not like owners in all ways or to the same degree (H3). Importantly, the categories represented in our dependent variable (“does not own and would not consider,” “does not own but would consider,” and “owns”) imply an inherent ordering that is well-suited to an ordered logistic regression framework. However, while the assumption underlying ordinal logit models is that the effects of the independent variable are constant, or proportional, across different thresholds of the dependent variable, previous research (Long and Freese, 2006) and our theoretical expectations suggest this may not always be the case. More specifically, if considerers are distinct from rejectors but not like owners in all ways or to the same degree, then we might expect certain variables to vary across thresholds of the dependent variable, rather than uniform relationships for each group. Because we cannot assume these differences *a priori*, we use Brant tests to formally test whether the proportional odds assumption holds.

Brant test results confirm our expectation: In all three studies, Need for Chaos and Stock Owner violate the assumption. Operational Ideology and Supports Developing AI violate the assumption in Study 2 and 3. White race violates the assumption in Study 1. Need for Cognition, and Investing Risk violate the assumption in Study 2. Trust in Government violates the assumption in Study 3. These results justify estimating a generalized ordered logit, or partial proportional odds model such that we can maintain the proportional odds assumption for explanatory variables that do not violate the assumption and relax it where the assumption fails (Williams, 2006; 2016). This approach helps us to avoid modeling strategies that violate the proportional odds assumption or disregard the ordinality of our dependent variable (as would be the case for ordered logit and multinomial logit, see Williams, 2006; 2016).¹¹

Following Williams (2006, 2016), we specify the generalized ordered logit, or partial proportional odds, model as:

$$P(Y_i > j) = \frac{\exp(\alpha_j + X_i \beta_j)}{1 + [\exp(\alpha_j + X_i \beta_j)]}, j = 1, 2, \dots, M - 1 \quad (1)$$

where Y_i represents the dependent variable, X_i represents the independent variables, α_j represents the intercept, β_j represents the estimated parameter, and M represents the maximum value of the dependent

⁹ We list alternative cryptocurrencies and comments included in open-ended responses in Appendix E in Supplementary Material.

¹⁰ Results from multivariate logit models for considerers (compared to rejectors) and owners (compared to all non-owners) are similar to results from bivariate models. Figures depicting bivariate relationships and a table of results corresponding with Fig. 5 are included in Appendices F - G in Supplementary Material.

¹¹ As an additional pre-test, we estimated an ordinary least squares regression model including all independent variables and variance inflation factors (VIF) for each study. The mean VIF was 1.35 in Study 1, 1.35 in Study 2, and 1.45 in Study 3, indicating multicollinearity was not a major concern.

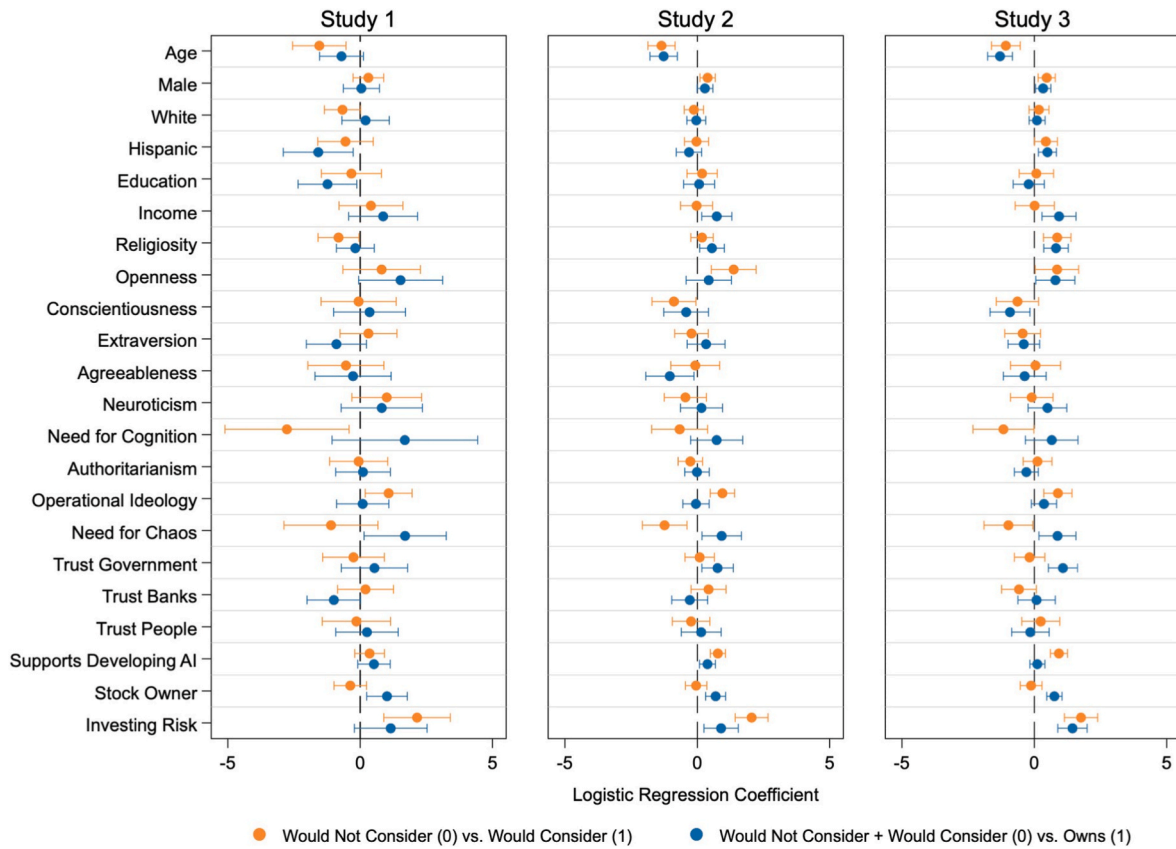


Fig. 5. Multivariate logit coefficient estimates for models estimating the probability of “would not consider” (coded as 0) versus “would consider” (coded as 1) and “would not consider” combined with “would consider” (i.e., all non-owners, coded as 0) versus “owns” (coded as 1). Horizontal error bars represent 95% confidence intervals, two-tailed tests. Models comparing “would not consider” to “would consider” only include each study’s subset of non-owners ($n = 695, 1,587,$ and $1,312$, in Study 1, 2, and 3, respectively), whereas those comparing “would not consider” + “would consider” to “owns” include each study’s full analytic sample ($n = 803, 1,908,$ and 2086 in Study 1, 2, and 3, respectively).

variable. We therefore specify the probability of each threshold of the dependent variable Y_i as:

$$\Pr(Y_i = 1) = 1 - g(X_i\beta_1) \quad (2)$$

$$\Pr(Y_i = j) = g(X_i\beta_{j-1}) - g(X_i\beta_j), \quad j = 2, \dots, M - 1 \quad (3)$$

$$\Pr(Y_i = M) = g(X_i\beta_{M-1}) \quad (4)$$

Because $M = 3$ in our case, we can interpret results from these models as a series of binary logistic regression coefficients, for which the categories of the independent variable are collapsed into two, simultaneously modeled groups: the probability of crossing the threshold from “would not consider” to “would consider,” then from “would consider” to “owns.” While variables that meet the proportional odds assumption should yield similar coefficients in both categories, those that violate the assumption may yield coefficients that vary in direction and/or strength (Williams, 2006, 2016). Fittingly, this approach allows us to test our three hypotheses simultaneously: that owners differ from non-owners, including considerers (H1), that considerers are distinct from rejectors (H2), and that, to the extent that considerers are distinct from rejectors, considerers are not like owners in all ways or to the same degree (H3). We present results from the generalized ordered logit, or partial proportional odds, models for each of the three studies in Table 1.

Variables that meet the proportional odds assumption are similar to logistic regression results for owning (shown previously in Fig. 5). Younger Age, greater Income, greater Openness, lesser Conscientiousness, greater Religiosity, and greater Investing Risk are positively associated with considering and owning in two or more studies. Male is also positively associated with both thresholds in two of three studies, suggesting

previous logistic regression models comparing owners to all non-owners may have inadvertently masked this effect. Results for Hispanic remain mixed across models but are significant and positively associated with both thresholds in Study 3.

Results for variables that violate the proportional odds assumption also help to refine earlier logistic regression model results. When estimated simultaneously, we observe that more conservative Operational Ideology (as indicated by the positive coefficient) and greater Supports Developing AI are more strongly related to considering than owning in two of three studies. Conversely, identifying as a Stock Owner and greater Need for Chaos are more strongly related to owning than considering in two or more studies.

Lastly, two variables exhibit mixed patterns: Greater Investing Risk significantly increases the probability of considering and owning across all studies but does not meet the proportional odds assumption in Study 2. Greater Trust Government significantly increases the probability of considering and owning in two studies (Study 2 and Study 3) but does not meet the proportional odds assumption in Study 3.¹²

¹² As an additional check, we ran a series of ordinal ordered logit models to assess whether the generalized ordered logit model yielded substantively different results (see Appendix H in Supplementary Material). As expected, this model specification consistently overestimates key variables (e.g., Operational Ideology) that the generalized ordered logit model indicates are more strongly related to considering than owning, while underestimating key variables (e.g., Need for Chaos) that the generalized ordered logit model indicates are more strongly related to owning than considering. These discrepancies further indicate that a generalized ordered logit, or partial proportional odds, model specification is the more appropriate approach for modeling these data.

To visualize these patterns more clearly, we plot results from Table 2 in Fig. 6. The orange circle and corresponding orange horizontal line indicate each independent variable's coefficient estimate and 95% confidence interval at the first threshold, i.e., the probability of considering cryptocurrency, as opposed to rejecting it (models 1a, 2a, and 3a in Table 2). The hollow blue circle and corresponding blue horizontal line represent each independent variable's coefficient estimate and 95% confidence interval at the second threshold, i.e., the probability of owning cryptocurrency, as opposed to considering (models 1b, 2b, and 3b in Table 2). Because both thresholds are modeled simultaneously (similar to an ordered logit framework, but with the added advantage that it can relax the proportional odds assumption where necessary), we can compare estimates in each group. Perfectly overlapping coefficient plots (such as those consistently shown for Age) indicate where a given variable operates uniformly for considerers and owners, whereas diverging coefficient plots (such as those consistently shown for Need for Chaos) indicate where a given variable operates differently for

considerers and owners. Again, despite some differences across the three independent samples, Fig. 6 helps to illustrate the more general pattern that owners are distinct from all non-owners (H1) and considerers are distinct from rejectors (H2), but considerers are not like owners in all ways or to the same degree (H3).

4.5. Average marginal effects

Beyond identifying which factors help to differentiate rejectors, considerers, and owners, we are also concerned with the substantive size of these predicted effects. Given logistic regression coefficients cannot be directly interpreted by size and significance alone, we clarify how key variables relate to each group by computing average marginal effects (AMEs). For every variable we present, the AME reflects the change in predicted probability of being in a given adoption category when that variable moves from its minimum to maximum values, holding all else constant. These predicted probabilities are then averaged across all

Table 2
Generalized ordered logit (partial proportional odds) model results.

	Study 1		Study 2		Study 3	
	(1a)	(1b)	(2a)	(2b)	(3a)	(3b)
Age	−1.30*** (0.37)	−1.30*** (0.37)	−1.43*** (0.20)	−1.43*** (0.20)	−1.31*** (0.20)	−1.31*** (0.20)
Male	0.16 (0.24)	0.16 (0.24)	0.36** (0.12)	0.36** (0.12)	0.44*** (0.12)	0.44*** (0.12)
White	−0.48 (0.30)	0.40 (0.46)	−0.08 (0.14)	−0.08 (0.14)	0.13 (0.13)	0.13 (0.13)
Hispanic	−0.98* (0.46)	−0.98* (0.46)	−0.20 (0.19)	−0.20 (0.19)	0.50*** (0.15)	0.50*** (0.15)
Education	−0.57 (0.41)	−0.57 (0.41)	0.15 (0.23)	0.15 (0.23)	−0.13 (0.24)	−0.13 (0.24)
Income	0.50 (0.46)	0.50 (0.46)	0.47* (0.22)	0.47* (0.22)	0.61* (0.27)	0.61* (0.27)
Religiosity	−0.60* (0.29)	−0.60* (0.29)	0.40* (0.17)	0.40* (0.17)	0.94*** (0.20)	0.94*** (0.20)
Openness	1.19* (0.56)	1.19* (0.56)	0.93** (0.33)	0.93** (0.33)	0.87** (0.31)	0.87** (0.31)
Conscientiousness	0.09 (0.52)	0.09 (0.52)	−0.62* (0.30)	−0.62* (0.30)	−0.91** (0.30)	−0.91** (0.30)
Extraversion	−0.26 (0.41)	−0.26 (0.41)	0.07 (0.26)	0.07 (0.26)	−0.38 (0.26)	−0.38 (0.26)
Agreeableness	−0.49 (0.60)	−0.49 (0.60)	−0.59 (0.34)	−0.59 (0.34)	−0.18 (0.34)	−0.18 (0.34)
Neuroticism	1.04 (0.57)	1.04 (0.57)	−0.21 (0.29)	−0.21 (0.29)	0.28 (0.30)	0.28 (0.30)
Need for Cognition	−0.84 (0.99)	−0.84 (0.99)	−0.09 (0.45)	0.75 (0.49)	−0.05 (0.45)	−0.05 (0.45)
Authoritarianism	0.02 (0.40)	0.02 (0.40)	−0.17 (0.18)	−0.17 (0.18)	−0.19 (0.20)	−0.19 (0.20)
Operational Ideology	0.78* (0.35)	0.78* (0.35)	0.57** (0.19)	0.12 (0.23)	0.83*** (0.22)	0.31 (0.23)
Need for Chaos	−0.24 (0.65)	1.72* (0.88)	−0.21 (0.33)	1.02** (0.35)	0.01 (0.31)	0.75* (0.32)
Trust Government	0.26 (0.48)	0.26 (0.48)	0.55* (0.23)	0.55* (0.23)	0.52* (0.24)	1.11*** (0.26)
Trust Banks	−0.33 (0.39)	−0.33 (0.39)	0.05 (0.25)	0.05 (0.25)	−0.08 (0.28)	−0.08 (0.28)
Trust People	0.07 (0.48)	0.07 (0.48)	0.01 (0.28)	0.01 (0.28)	−0.01 (0.29)	−0.01 (0.29)
Supports Developing AI	0.44* (0.22)	0.44* (0.22)	0.69*** (0.12)	0.35* (0.15)	0.64*** (0.13)	0.16 (0.14)
Stock Owner	0.12 (0.25)	0.88* (0.37)	0.24 (0.16)	0.69*** (0.18)	0.28 (0.14)	0.77*** (0.14)
Investing Risk	1.82*** (0.51)	1.82*** (0.51)	1.77*** (0.25)	1.09*** (0.30)	1.74*** (0.23)	1.74*** (0.23)
Constant	−1.18 (0.83)	−4.00*** (0.95)	−1.55*** (0.46)	−3.00*** (0.47)	−1.67*** (0.43)	−2.96*** (0.44)
n	803		1908		2086	

Note: Table values are generalized ordered logit (partial proportional odds) coefficients with standard errors in parentheses. Coefficients in models 1a, 2a, and 3a can be interpreted as those from a binary logit regression where the dependent variable is coded as “would not consider” vs. “would consider” combined with “owns.” Coefficients in models 1b, 2b, and 3b can be interpreted as those from a binary regression where the dependent variable is coded as “would not consider” combined with “would consider” vs. “owns.” * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

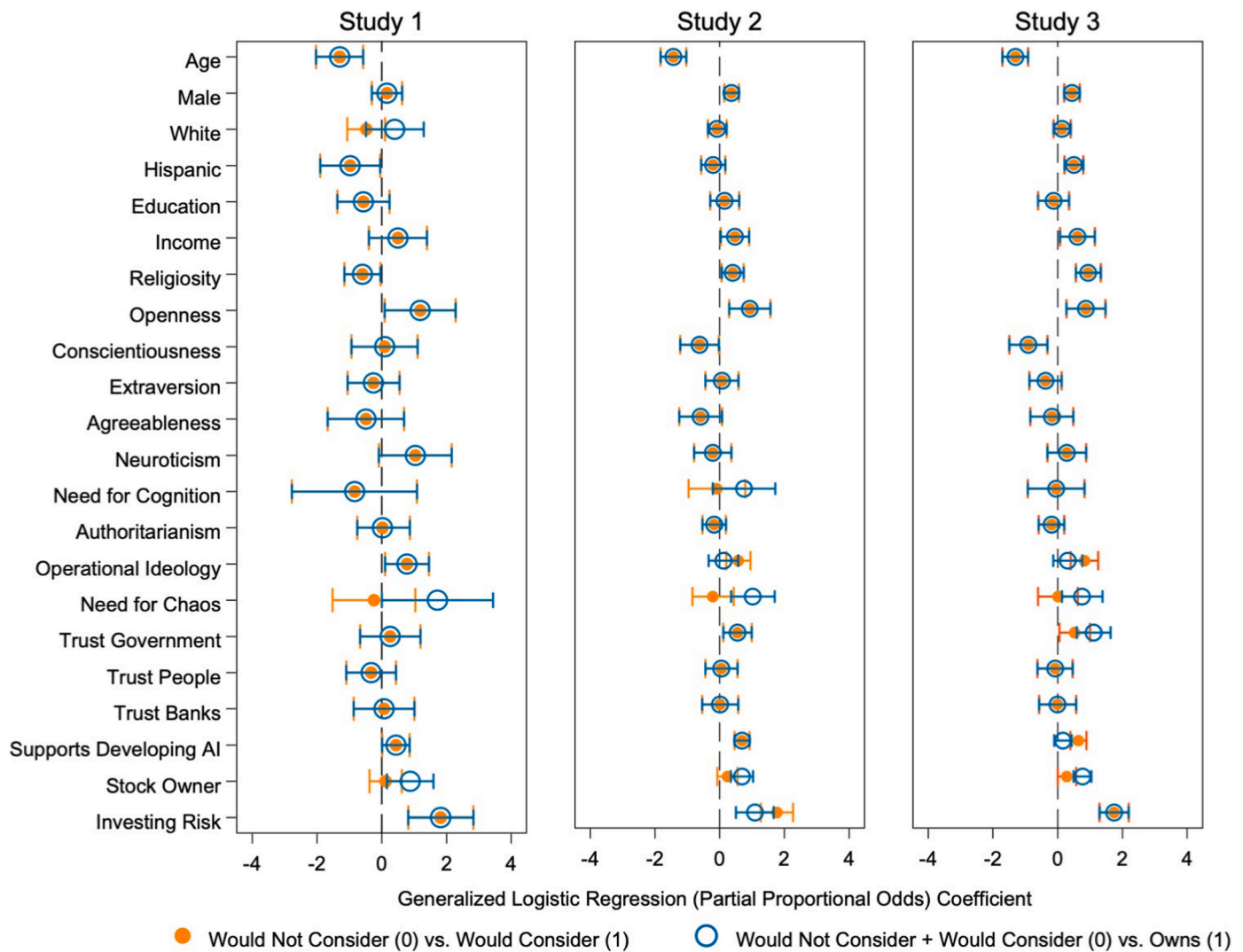


Fig. 6. Generalized ordered logit (partial proportional odds) model results corresponding with Table 2. Orange circles and orange horizontal lines indicate logistic regression coefficient estimates and 95% confidence intervals (two-tailed tests) for independent variables at the first threshold, i.e., the probability of considering cryptocurrency, as opposed to rejecting (models 1a, 2a, and 3a in Table 2). Hollow blue circles and blue horizontal lines indicate logistic regression coefficient estimates and 95% confidence intervals (two-tailed tests) for independent variables at the second threshold, i.e., the probability of owning cryptocurrency, as opposed to considering (models 1b, 2b, and 3b in Table 2). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

respondents. Positive values indicate increases in the probability of being in that category; negative values indicate decreases in the probability of being in that category. AMEs therefore allow us to determine not only which category of the cryptocurrency adoption process is affected, but how strongly each variable relates to a given category. For brevity, we focus on variables that yield the most consistent results across the three independent samples and calculate AMEs based on values from Study 3. We provide these results in Table 3 (AMEs for Study 1 and Study 2 are included in Appendix I of Supplementary Material).

For variables that meet the proportional odds assumption, AMEs suggest that they operate in a consistent, ordered manner across both thresholds, i.e., these variables are significantly associated with a change in the probability of considering and owning. The observed changes for considering and owning are also in the same direction and are largely the result of being more or less likely to reject. For example, varying *Age* from the minimum to maximum in our data, the oldest respondents are 24 percentage points more likely than the youngest respondents to reject the possibility of owning cryptocurrency, while being 4 percentage points less likely to consider cryptocurrency and 20 percentage points less likely to own. In contrast, those who score highest

in *Investing Risk* are 32 percentage points less likely than their lowest-risk counterparts to reject the possibility of owning cryptocurrency, while being 5 percentage points more likely to consider cryptocurrency and 27 percentage points more likely to own.

For variables that violate the proportional odds assumption, AMEs help to further emphasize differences between groups. More specifically, these observed changes in probability are not necessarily in the same direction and may result from being more or less likely to reject or consider. For example, varying *Operational Ideology* from the most liberal to most conservative values in our data increases the probability of considering (as opposed to rejecting) by 10 percentage points, but does not increase the probability of owning. In contrast, varying *Need for Chaos* from the minimum to maximum values in our data does not change the probability of considering versus rejecting, but does increase the probability of owning versus considering by 11 percentage points.

In terms of which variables have the strongest effects, *Investing Risk* yields the highest change in predicted probability, both in terms of being less likely to reject cryptocurrency and more likely to own. Also noteworthy are the AMEs for *Age*, *Religiosity*, *Openness*, and *Conscientiousness*, which previous research also found to have comparable marginal effects

Table 3

Average marginal effects.

	Would Not Consider	p- value	Would Consider	p- value	Owens	p- value
<i>Meets the PO Assumption</i>						
Age	0.24	0.000	−0.04	0.000	−0.20	0.000
Male	−0.08	0.000	0.01	0.002	0.07	0.000
Income	−0.11	0.027	0.02	0.034	0.09	0.027
Religiosity	−0.17	0.000	0.03	0.000	0.14	0.000
Openness	−0.16	0.004	0.03	0.008	0.13	0.005
Conscientiousness	0.17	0.002	−0.03	0.004	−0.14	0.003
Investing Risk	−0.32	0.000	0.05	0.000	0.27	0.000
<i>Violates the PO Assumption</i>						
Operational Ideology	−0.15	0.000	0.10	0.001	0.05	0.171
Need for Chaos	−0.00	0.981	−0.11	0.001	0.11	0.017
Supports Developing AI	−0.12	0.000	0.09	0.000	0.02	0.243
Stock Owner	−0.05	0.051	−0.07	0.001	0.12	0.000

Note: Average marginal effects estimated from generalized ordered logit models in Study 3. AMEs for Study 1 and Study 2 are included in [Appendix 1](#) in Supplementary Material.

(Ferguson et al. 2024). The effects of *Need for Chaos* and *Stock Owner* for owning cryptocurrency are similarly strong, followed by the effect of *Operational Ideology* for considering. *Male* and *Income* (for owning) and *Supports Developing AI* (for considering) yield the smallest AMEs, consistent with the idea that certain factors may shape behavior indirectly but may matter less once positive considerations and intentions are formed (Ajzen, 2020).

5. Discussion

Using three original, nationally-representative, cross-sectional samples of US adults ($n = 803$; $n = 1908$; $n = 2086$), we examine which factors are associated with rejecting, considering, and owning cryptocurrency in the United States. Across three identical analyses (each using one independent sample), we find that owners differ from non-owners (H1) and considerers differ from rejectors (H2); however, considerers are not like owners in all ways or to the same degree (H3). While factors like age, gender, and openness to experience relate to both considering and owning, more conservative operational ideology and support for developing AI are more strongly associated with considering than owning. At the same time, stock ownership and Need for Chaos are more strongly associated with owning than considering. Together, these patterns build upon existing behavioral theories to demonstrate that cryptocurrency “considerers” are a distinct and meaningful group.

In turn, our work challenges common assumptions underlying earlier cryptocurrency adoption research. More specifically, our findings suggest that distinguishing between rejectors, considerers, and owners reveals meaningful differences that are likely obscured under a binary ownership model. We find that variables previously dismissed as inconsequential (e.g., operational ideology) do, in fact, matter—just not necessarily at the ownership stage. Likewise, we find that disruptive orientations (e.g., Need for Chaos) primarily arise at the ownership, but not the considering, stage. We further demonstrate that investigating these differences simultaneously in multivariate models is necessary to identify how a certain segment of a population is related to a given adoption stage, all else equal. In turn, our work demonstrates that cryptocurrency adoption is not monolithic. Therefore, accounting for heterogeneity among non-owners is analytically and practically important for researchers, policymakers, and communicators aiming to understand who they might most effectively study, target, or persuade.

Additionally, our results reaffirm well-established findings, e.g., younger individuals and males are more likely to own cryptocurrency. However, results from all three independent samples indicate that these

characteristics may not yield the strongest predicted effects. In our data, investing risk consistently yields the largest average marginal effect, reducing rejection and increasing ownership, all else equal. Furthermore, although the link between investing risk and being more likely to consider and/or own cryptocurrency may seem obvious, it bears repeating that risk-taking reflects perceived, rather than objective, risk (see, e.g., Weber et al., 2002), and may therefore speak to the notion of control beliefs emphasized in the Theory of Planned Behavior (TPB). If so, then this would suggest that the strongest determinant of actually owning cryptocurrency is perceived behavioral control vis-à-vis financial risk, i.e., that individuals must believe cryptocurrency is both worth considering and realistically within their capabilities to adopt.

Results for operational ideology, Need for Chaos, support for developing AI, and stock ownership also fit into this framework. The relationship between more conservative operational ideology and considering, as evidenced by our data, reaffirms that background factors and attitudes alone do not propel individuals toward behavior without sufficient motivation and control beliefs (Ajzen, 1985, 2020). Previous experimental work on economic decision making further suggests that Need for Chaos is not merely a different form of risk tolerance, but a disruptive, gain-focused orientation that can motivate systematic differences in behavior (Farhart et al., 2023; Fitz et al., 2025; Petersen et al., 2023). That support for developing AI yields the smallest significant effect (second only to being male and also comparable to reporting higher income) on increasing one's probability of considering, rather than rejecting, is also in line with the idea that certain factors may shape behavior indirectly but often matter less once positive considerations and intentions are formed (Ajzen, 1985, 2020). Lastly, it may also be the case that stock owners, who already engage in speculative financial markets, possess stronger perceptions of behavioral control, which then facilitates behavioral follow through. In TPB terms, the implication of our findings is that prior investment experience may make cryptocurrency ownership appear to be a more actionable choice.

5.1. Theoretical implications

While we recognize that the Theory of Planned Behavior (TPB) may not fully encapsulate the complexity of individual-level adoption decisions, it remains a useful framework for articulating the core assumption that consideration precedes action. Other attitude-behavior models that rely on additional assumptions (e.g., habitual behavior and behavior in social contexts, see, e.g., Triandis, 1977; Gibbons et al., 1991) may help to further expand upon the factors associated with cryptocurrency adoption, whereas theories that emphasizes the role of social influence and perceptions of legitimacy at the system level (e.g., diffusion theory, see Rogers, 2003) may provide additional insights into how adoption evolves.

Situating our individual-level data in the system-level diffusion theory model, for example (see Fig. 7), suggests that despite sample differences in self-reported ownership rates (13% in Study 1, 18% in Study 2, and 32% in Study 3), cryptocurrency adoption in the United States is in or nearing the early majority phase. While it remains to be seen whether cryptocurrency follows the conventional adoption curve, one implication of our findings is that the motivations for cryptocurrency adoption are stratified, as some individuals approach it as a technological innovation, others as a financial asset, and others as a symbolic or oppositional tool. We look forward to research that explores these complementary perspectives.

Another possibility is that consideration reflects those who do not reject cryptocurrency outright. At the same time, both types of non-owners (i.e., rejectors and considerers) may represent groups more susceptible to disruptions and elite cues. For example, although liberal ideology has historically been associated with greater electric vehicle (EV) adoption (Sintov, 2020), perceptions of Elon Musk have increasingly moderated liberals' EV adoption intentions (Flores et al., 2025). Psychological adaptations like Need for Chaos may also emerge after

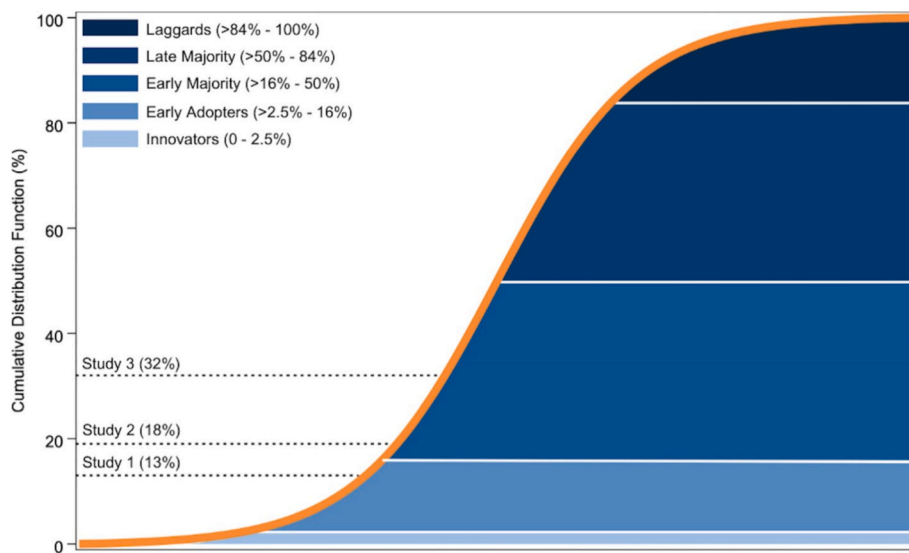


Fig. 7. An adaption of Rogers's (2003) diffusion of innovations curve, shown as an s-curve (i.e., cumulative distribution function, or CDF, as scaled on the left-side y-axis indicated by the solid orange line). Blue segments under the s-curve represent probability shading for each of the five adopter groups in Rogers's model, based on standard deviations (σ) from the mean (μ) in a normal distribution (the percent ranges for which are shown in parentheses next to each adopter group name). In a normal distribution, these segments represent how much of the total area (probability) lies within that standard deviation range: Innovators represent 2.5% ($0, \mu - 2\sigma$), Early Adopters represent 13.5% ($\mu - 2\sigma, \mu - 1\sigma$), Early Majority represents 34% ($\mu - 1\sigma, \mu$), Late Majority represents 34% ($\mu, \mu + 1\sigma$), and Laggards represent 16% ($\mu + 1\sigma, 1$). Black dotted horizontal lines indicate self-reported cryptocurrency ownership rates in our three studies (13% in Study 1, 18% in Study 2, 32% in Study 3). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

adoption if owners gravitate towards disruptive sentiments to reaffirm their adoption decision (see, e.g., Ajzen, 2020; Greenhalgh et al., 2005). While we cannot speak to the temporal ordering of these relationships, our findings suggest that distinguishing between considering and owning helps to better capture potential differences across sociopolitical, psychological, technological, and financial domains.

Our measurement and modeling approaches further suggest that it is possible for researchers to capture meaningful variation between heterogeneous groups without relying on an exhaustive (and potentially noisy) set of items capturing perceptions, intentions, and control beliefs. While we do not propose that this strategy is exhaustive nor applicable to every behavior, we are confident that it offers a valid starting point for studying the prerequisite considering stage.

5.2. Managerial implications

From a managerial perspective, our findings underscore the importance of treating “considerers” as a meaningful group. Although the Theory of Reasoned Action (the predecessor of the TPB) posited that intentions could operate as proxies for behavior “[w]hen people have perfect volitional control,” the TPB helped to establish that decision making is not fully rational and intentions do not always predict whether behavior occurs (Ajzen, 2020, 316). Understanding what, when, and for whom certain factors matter for intentions and behavior may be especially important for a decentralized, volatile, and evolving product like cryptocurrency, suggesting those involved in decision making should distinguish between rejecting, considering, and owning groups. While this is not meant to suggest that considerers will always differ from rejectors and/or owners in every domain, our data (and the TPB) suggest that accounting for considerers is necessary to determine whether such differences exist. Moreover, failing to account for considerers risks obscuring meaningful variation among non-owners, misidentifying target audiences, and ultimately directing strategies toward the wrong people and/or levers of influence.

Understanding who rejects cryptocurrency may further help to signal the various barriers to adoption, providing valuable guidance for how to design effective strategies to influence awareness, knowledge, and

perceived risk. At the same time, our findings highlight that the crypto-engaged are not a single audience, and strategies that treat them as such risk overlooking key differences between groups. For example, messaging and services aimed at promoting legitimacy and reducing uncertainty may encourage considerers but backfire among existing owners, who may value risk-taking, novelty, and the notion of cryptocurrency as an oppositional tool (see, e.g., Greenhalgh et al., 2005). Conversely, messaging emphasizing exclusivity, sophistication, or financial autonomy may resonate with owners but discourage considerers, who may be more sensitive to subjective norms and negative control beliefs. These tensions may be particularly relevant for an emerging, rapidly evolving, and increasingly politicized market like cryptocurrency (see, e.g., Fig. 8), where decentralized networks and peer communities largely substitute for conventional oversight but advocates, firms, and service providers may still need to balance both mainstream and alternative frames.

5.3. Limitations and directions for future research

Despite consistent results across multiple samples, we highlight three qualifications to our findings: First, we rely on cross-sectional observational data, such that we can only speak to the consistency of our findings across multiple samples, but cannot establish causality, the stability of considering, or how people might progress through adoption stages over time. Second, we acknowledge that there may be additional heterogeneity among considerers, who may share in some general openness to adoption but vary in interest, knowledge, and intentions to act. Our approach is not meant to suggest that “considering” represents a fixed or uniform stage, but rather that considering is an evaluative orientation—a meaningful point of behavioral willingness that, according to the Theory of Planned Behavior, precedes intention and behavior, and relates more directly to the background factors elucidated in previous ownership research. While we lack panel data required to test individual-level change over time, our findings provide strong evidence as to how people might think about cryptocurrency and make adoption decisions. We look forward to future longitudinal research that continues to build on this work.



Fig. 8. Truth Social post by @realDonaldTrump about his own digital currency, \$TRUMP, on March 23, 2025. <https://truthsocial.com/@realDonaldTrump/114212473050636851>

We further acknowledge that, despite accounting for a wide range of variables, there may still be unobserved factors related to considering and owning not captured in our data. Our study is also limited to US adults; therefore, we look forward to future studies that test the relationships observed in our study in the context of other social, political, and/or economic systems. Lastly, we draw the readers' attention back to the fact that, despite rigorous attention to survey design and representative samples, our samples vary in self-reported ownership rates. To better understand these discrepancies, we conducted a series of Oaxaca–Blinder decompositions (see Appendix J in Supplementary Material), which indicate that the observed differences in ownership rates are partially attributable to sample differences in age.¹³ While our main findings suggest that age is an important, but not the strongest, predictor of considering and owning, we look forward to future research that continues to explore how age relates to the cryptocurrency adoption process. For example, researchers might examine whether younger individuals are more exposed and responsive to technological innovations downstream of internet use and chosen media sources (see, e.g., Littrell et al., 2024). The relationship between age and ownership may also reflect social desirability bias, as individuals over- or under-report ownership depending on how socially acceptable or politically-charged they perceive cryptocurrency to be.

Additionally, while it may be tempting to translate our findings into prescriptive guidance, doing so (e.g., providing detailed policy recommendations or other organizational strategies) would require additional assumptions beyond what our observational data can support. We also know relatively little about for whom, how, and when the relevant factors revealed in our data might interact—questions that are notably central to the TPB. Identifying which of these attributes *cause* individuals to consider and/or own cryptocurrency in various contexts would require experimental designs (e.g., message framing experiments or platform- or product-specific A/B tests). We also do not know how individuals conceptualize what cryptocurrency “is” and the range of crypto-related applications far exceed any one sector alone. In turn, our findings should not be interpreted as explanatory or a turnkey solution, but as a guiding framework that distinguishes between rejectors, considerers, and owners.

Because these findings align with decades of attitude-behavior research, however, we would expect them to be broadly applicable even amid variation in how cryptocurrency is understood and/or used. Regardless of how the cryptocurrency landscape evolves, the implication of our findings is that both consideration and ownership reflect myriad individual differences, including (but not limited to) perceived utility, social acceptability, value alignment, and control. As such, what drives attitudes, beliefs, and behavior may be as much about meaning as it is about the innovation itself.

¹³ It is important to note that while we could examine differences in ownership rates for any of the variables in our analyses, we focus on Age because it is one of the only variables that noticeably differs in the three independent samples.

Lastly, some readers may question whether our findings are specific to certain types of cryptocurrency (e.g., Bitcoin) rather than cryptocurrency, broadly defined. While consideration and ownership can, in principle, be disaggregated by asset type, doing so would require purposely restricting our samples in ways that are contradictory to best practices and, in turn, our study approach. Recall that, in our data, the types of cryptocurrencies respondents report owning closely mirror broader market patterns, with Bitcoin constituting the majority share. However, a sizeable portion of respondents report owning other types of cryptocurrency and many report holding multiple types (see again Section 4.2). In other words, many owners report holding Bitcoin *in addition to* other cryptocurrencies, rendering it virtually impossible to disentangle asset-specific relationships in a meaningful or reliable way.¹⁴ More fundamentally, understanding this phenomenon requires assessing whether the factors associated with particular types of cryptocurrency are indeed distinct to that asset or asset type, rather than reflecting broader attitudes and/or behaviors (see, e.g., Ferguson et al., 2023, who compared cryptocurrency owners to stock owners). Once ownership has been measured in a way that adequately captures these possibilities, selectively discarding data is not a defensible analytic choice. Avoiding these and other pitfalls was a central premise of the present study, and we hope our approach encourages similar attention to the issues of selection and stage-specific inference in future work.

CRedit authorship contribution statement

Erin B. Fitz: Writing – original draft, Visualization, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.
Kyle L. Saunders: Writing – original draft, Visualization, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing interests.

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¹⁴ For example, in Study 3, the majority of owners (57%) reported holding Bitcoin alongside other cryptocurrencies, while another sizeable portion (13%) reported holding no Bitcoin at all.

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Supplementary Material

Supplementary material for this article can be found online at <https://doi.org/10.1016/j.jretconser.2026.104759>.

Data availability

Data will be made available on request.

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